ZYAD MOHAMED ABDELFATTAH

GIS Developer | GIS Analyst | ArcGIS Pro · Python/ArcPy · AutoCAD & Civil 3D

Kafr El Dawar, Beheira, Egypt | +20 102 537 0757 | +20 120 431 9579 | z7887008@gmail.com
LinkedIn: linkedin.com/in/zyad-mohamed-abdelfattah-37b9223bb | Portfolio: zyad-mohamed-abdelfattah.vercel.app | GitHub: github.com/zyadweb4

PROFESSIONAL SUMMARY

GIS Developer and GIS Analyst with a B.A. in Geography (Surveying & GIS Division, Damanhour University, 2026) and about four years of hands-on work in ArcGIS Pro, QGIS, AutoCAD and Civil 3D across field surveying, technical-office and academic projects. Developer of **GeoPilot**, an ArcGIS Pro add-in built with ArcGIS Pro SDK, C#/.NET 8, WPF and a Python/ArcPy engine that automates hydrology, land subdivision, sewer and water network design, survey data import and map production. Experienced in spatial analysis, geodatabase management, georeferencing, coordinate systems, CAD-to-GIS conversion and GIS workflow automation. Completed Esri's Spatial Data Science MOOC and ITI's 60-hour Introduction to ArcGIS in 2026.

CORE SKILLS

GIS Software: ArcGIS Pro, ArcGIS Desktop (ArcMap), ArcGIS Online, QGIS, Global Mapper, ModelBuilder, Spatial Analyst, Network Analyst, 3D Analyst

Spatial Analysis: hydrological analysis (DEM, flow direction, flow accumulation, stream order, watershed delineation), buffer and proximity analysis, nearest neighbor analysis, spatial distribution and dispersion (standard distance, directional distribution), urban extent and urban growth mapping, suitability analysis, network and service-area analysis, terrain analysis (slope, aspect, hillshade, cut and fill, viewshed), hot spot analysis, spatial statistics

Editing & Queries: digitizing and feature editing (Editor tools), attribute queries and definition queries, Select by Attributes, Select by Location, coordinate import and export, coordinate extraction from drawings

GIS Development & Automation: Python, ArcPy, Python toolboxes, ArcGIS Pro SDK, C#, .NET 8, WPF, Arcade attribute rules, geoprocessing automation, pyproj, pytest, Git and GitHub

Data Management: file geodatabase design, feature classes and domains, topology and data quality control, georeferencing, digitizing, coordinate reference systems and projections (UTM 35N/36N, WGS 84), SQL queries

Remote Sensing: Sentinel-2 and Landsat imagery, NDVI, vegetation and water cover analysis, change detection, land suitability analysis, deep-learning object detection in ArcGIS Pro

Surveying & CAD: Total Station, GPS/GNSS, Level, Echo Sounder, Sokkia Link, setting out and coordinate stake-out; AutoCAD (certified) 2D drafting and sheet layouts, Civil 3D (surfaces, 3D road corridors, pipe networks, sewer and drainage design, profiles, cut and fill, quantity take-off/BOQ), InfraWorks and SketchUp 3D modeling, Surfer; DEM and contour generation

Data Formats: Shapefile, File Geodatabase, GeoJSON, KML/KMZ, DWG/DXF, LandXML, PNEZD, CSV/Excel, LAS point clouds, EPANET .inp

Office & AI Tools: Microsoft Excel, Word, PowerPoint, Power BI; Claude, Claude Code, ChatGPT, Gemini

PROFESSIONAL EXPERIENCE

GIS Developer | GeoPilot — ArcGIS Pro Add-in **2026 – Present**
Independent project | ArcGIS Pro SDK, C#/.NET 8, WPF, Python, ArcPy

- Designed and developed an ArcGIS Pro add-in with ribbon commands, a dockable bilingual (Arabic/English) panel, map tools and licensing, running a Python/ArcPy geoprocessing engine with 90+ automated actions.
- Built one-step analysis packages: flash-flood hazard (DEM conditioning, flow direction and accumulation, stream order, watersheds, morphometry, El-Shamy hazard classification), least-cost routing, solar site selection and smart land subdivision with land-budget tables.
- Developed engineering design tools: gravity sewer network design to the Egyptian Code (ECP 101) with invert levels, longitudinal profiles exported to DXF and BOQ; water network analysis and pipe sizing validated against EPANET 2 (Net1, heads within 0.011 ft).
- Implemented data conversion inside ArcGIS Pro: DWG/DXF to and from GIS, Excel/CSV to map, LandXML and PNEZD exchange with Civil 3D, KML/KMZ, GeoJSON, Shapefile, and survey instrument files (Leica GSI, Trimble JobXML, Sokkia SDR33) with field-to-finish coding.
- Added automatic Arcade attribute rules (14 calculation and 10 validation rules), 6 one-click project templates, a library of 128 ready ArcPy scripts and print-ready map layouts with branded PDF reports.
- Covered the engine with 398+ automated tests and documented every action with inputs, outputs and limits.

QA Engineer (Web & Mobile) | Findo Group **Oct 2025 – Present**

Remote, Full-time

- Test web platforms and Flutter mobile applications (Android and iOS) before release; document reproducible issues with steps, expected and actual results, severity and screenshots, and track them in Jira.
- Applied the same data-quality mindset to GIS work: validating records, attributes and outputs before delivery.

Data Entry & SEO Specialist | Findo Group **Oct 2023 – Oct 2025**

Remote, Full-time

- Entered, validated and maintained bilingual (Arabic/English) business records and prepared reports in Excel, Word and PowerPoint.

SELECTED GIS PROJECTS

AI-Integrated Smart City — Hurghada, Red Sea (Graduation Project) **2023 – 2025**

- Ranked 1st in the Surveying & GIS Division for project execution; awarded a certificate of appreciation for excellence.
- Collected field data with Total Station, GPS Station, Level and Echo Sounder; processed coordinates and produced terrain models, contour maps and thematic maps in ArcGIS Pro, Civil 3D, AutoCAD and Surfer.
- Built an integrated GIS model of Makadi from 61,000+ survey points with terrain, roads, water, electricity and sewer layers, plus hydrological, topographic and service-area analyses on the Egyptian topographic grid.
- Designed a coastal sewer and stormwater network in Civil 3D with longitudinal and cross sections, profile sheets and a bill of quantities.

GIS, Technical Office & CAD Work **2022 – 2026**

- Digitized and edited spatial data in ArcGIS Pro, built attribute and location queries, and extracted and imported survey coordinates between field files, CAD drawings and GIS layers.
- Performed hydrological, buffer, nearest neighbor, dispersion and urban-extent analyses and produced thematic map layouts for each study.
- Prepared and oriented plot sheets in AutoCAD, and drafted 2D plans (grid axes, columns, rooms) that were then modeled in 3D in SketchUp.
- Designed 3D road corridors and sewer/drainage networks in Civil 3D with longitudinal profiles, cut and fill volumes and quantity take-off, and built 3D site models in InfraWorks.

gis-survey-toolkit — Open-source Python Toolkit (GitHub) **2026**

- Python toolkit that converts total-station and GPS survey points (UTM CSV) to GeoJSON and KML, transforms UTM to and from WGS 84 with pyproj, and computes parcel area (m² and feddan), perimeter, bearings and centroids.
- Command-line interface, input validation for field files, pytest test suite and GitHub Actions CI on Python 3.10 and 3.12.

EDUCATION

Bachelor of Arts in Geography — Surveying & GIS Division **2022 – 2026**

Faculty of Arts, Damanhour University, Egypt | Grade: Very Good | Ranked 3rd in the faculty

CERTIFICATIONS

- Spatial Data Science: The New Frontier in Analytics (MOOC, 6 weeks) — Esri, Nov 2026
- Introduction to ArcGIS (60 hours) — Information Technology Institute (ITI), Mar – Jul 2026
- Getting Started with Spatial Analysis — Esri, Nov 2026
- Working with Charts and Reports in ArcGIS Pro (ArcGIS Lab) — Esri, Feb 2025
- ArcGIS Pro Basics — Esri, Feb 2025
- Surveying Programs: Sokkia Link, Surfer, AutoCAD, Civil 3D (Grade: Very Good) — Misr Public Library, Damanhour, Nov 2024 – Jun 2025
- Statement of Proficiency in Surveying Instruments and GIS/CAD Software — Geography Department, Damanhour University, 2023 – 2025
- AutoCAD Certificate (professionally accredited)

LANGUAGES

Arabic: Native | English: Good (working proficiency)